Thermodynamics (NMEC103)
Mid Semester Examination
Monsoon Semester 2025-2026
Total Marks: 60

Date: 20^{th} February 2025 | Total Time: 2 Hours

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1. Answer all the questions. [2×8=16 Marks]

   (a) A fluid is passing through a nozzle steadily. One engineer claims that the change in specific kinetic energy can be expressed as $\Delta ke = -\int vdp$. Do you agree with this? If yes, clearly show the conditions.

   (b) Give an example of a thermodynamic process such that $\int pdv \neq 0$ but $w = 0$. Also, show a case where $\int pdv = 0$ but $w \neq 0$.

   (c) A closed stationary insulated system undergoes a process by irreversible paddle wheel work input. In another case, the same system undergoes a frictionless quasistatic moving boundary work input process between the same end states. One engineer claims that the magnitude of work interaction in the above two cases will differ. Do you agree? If yes, give justification.

   (d) $\delta q = du + pdv$. Is this expression valid for a steady flow process? If yes, clearly mention the assumptions.

   (e) What do you mean by simple compressible pure substance?

   (f) During the charging of an insulated tank, the temperature rises, and during discharging, it drops. Do you agree? If yes, provide the physical reasoning behind it.

   (g) Water and steam are present together in a rigid, closed container. The system is heated, and it is observed that there is no vapour in the container after some time; only liquid is present. Can this observation be actual? If yes, explain.

   (h) An ideal gas undergoes throttling across an adiabatic valve. Which, if any of the following quantities, is non-zero for the process? ΔU, ΔH, ΔT, ΔP, W? Justify.

2. The gas inside the piston-cylinder assembly in Fig. 1 has a volume of 0.1m^3 and a pressure of 100kPa. This pressure balances the weight of the piston exactly, such that there is no stress exerted on the spring. 130kJ of heat is transferred to the gas, changing its state to 300kPa and 0.2m^3. Assuming the spring to be linear, determine [8 Marks]

   (a) Draw this process on the $p - v$ plot.

   (b) Find the work done by or on the system (i.e., gas) and the percentage of work done on spring.

   (c) Find the changes in internal energy and enthalpy of gas.

   (d) Based on the result of part (c), what is the change in stored energy of the system?

$P = C_1 + C_2 V$
100kPa = C_1 ...
300 kPa

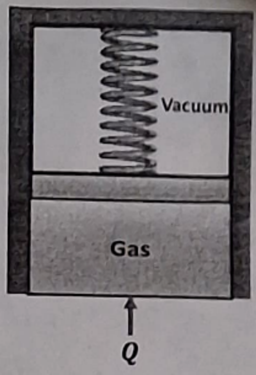

Figure 1: See Q. 2

3. As shown in Fig. 2, an airline 300K and 0.5MPa is connected to a turbine that exhausts to a closed initially empty tank of 50m^3. The turbine operates to a tank pressure of 0.5MPa, at which point the temperature is 250K. Assuming the entire process is adiabatic, determine the turbine work. Clearly state your assumptions. Assume air to be callorifically perfect gas with $C_p = \mathbf{1.005}\text{kJ/(kg.K)}$, $M = \mathbf{28}\text{kg/kmol}$. [6 Marks]

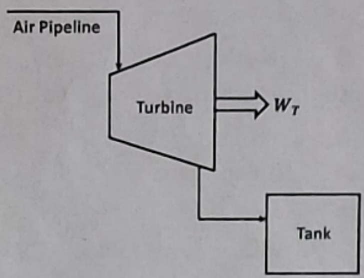

Figure 2: See Q. 3

4. Answer the following questions in brief.

   (a) Identify various components of the *total energy* and *internal energy* with a brief description of these. [2 Marks]

   (b) How *heat, internal energy, and thermal energy* are inter-related? [2 Marks]

   (c) What are the various forms of the *Mechanical Energy* of charged ionized plasma flowing under the influence of gravity and external electric and magnetic fields? [2 Marks]

   (d) Consider the falling of a boulder of mass M into a river and eventually settling at the bottom of the riverbed. Starting from the initial potential energy of the boulder, identify and illustrate the energy transfer processes involved during various stages of its fall and settling down. [2 Marks]

   (e) Do you recognize *specific enthalpy* of a flowing fluid as a form of *specific energy*? Clearly illustrate your arguments. [2 Marks]

   (f) Identify the *heat* and *work* interaction for a room, if

      i. The room is heated by an electric coil heater plugged into an external electric supply line. Take the entire room, including the heater, as a *system*. [2 Marks]

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